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SUB-SOIL INVESTIGATION REPORT

FOR THE PROPOSED CONSTRUCTION OF 10 STORIED BUILDING
AT GRAMEEN BANK, ZONAL OFFICE, MOUCHAK R/A, KALIPUR, SUNAMGONJ.
DAG # CS/SA-3240/3440, RS/BS-8551, KHA # CS/RS-289, RS/BS-8521, NAMJARI-289/1
J.L # CS/SA-144, RS/BS-95, MOUJA- TEGRIA

CLIENT

GRAMEEN BANK ZONAL OFFICE, SUNAMGONJ

Verification Code: egY4q6BJ
Verify report at: crtc-cee-sust.com

CONDUCTED BY

CENTER FOR RESEARCH TESTING & CONSULTANCY
DEPARTMENT OF CIVIL & ENVIRONMENTAL ENGINEERING
SHAHJALAL UNIVERSITY OF SCIENCE & TECHNOLOGY, SYLHET.

SAMPLE COLLECTED BY

M/S RELATION SOIL INVESTIGATION
Kumarpara, Sylhet.
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Introduction

A reasonably accurate conception about the sub-soil parameters of any project site is an essential prerequisite for proper planning and designing the foundation of the structure so that the same after its construction would remain safe and serviceable. A detailed study on the sub-surface elements in this regard would be necessary to select proper type of foundation.

Objectives

The purpose of sub-soil investigation program is to obtain the following information

1. To classify the formation of soil layers
2. To estimate the safe bearing capacities of the foundation at different soil layers
3. To suggest the safe and appropriate type of foundations.

Scope of work

Centre for research, testing and consultancy (CRTC), Department of Civil and Environmental Engineering, Shahjalal University of Science & Technology (SUST), Sylhet; execute only the laboratory tests and other computational works with the supplied disturbed and/or undisturbed samples to determine the geotechnical properties of the soil.

Table – 1: Items of the supplied samples

No	Name	Bore hole no	Depth	Quantities
01	Disturbed sample	01	0-135 ft	27
02	Disturbed sample	02	0-135 ft	27
03	Disturbed sample	03	0-135 ft	27
04	Disturbed sample	04	0-135 ft	27
05	Disturbed sample	05	0-135 ft	27
06	-	-	-	-
07	-	-	-	-
08	-	-	-	-
09	-	-	-	-
10	-	-	-	-

Laboratory Classifications

The disturbed soil samples, supplied by the client were thoroughly examined and classified roughly (color and type) and noted in accordance with bore hole (BH) and depth (Please see enclosed Tables).

Laboratory Works & Analysis

In addition to visual examination of the sample, Bearing Capacity of soil samples including skin friction & end bearing were calculated.

Bearing capacity for shallow foundation

The N value gives useful information with regards to consistency of cohesive soil and relative density of cohesion less soil. These relationships are given in the following Tables for cohesive and cohesion less soil, respectively.

Table 2: Relation of N-value with the Consistency and the Unconfined Compressive Strength of Cohesive Soils (after Bowles 1997)

N-values	Consistency	Unconfined Compressive Strength (q_u)	
		tsf	kPa
0-2	Very soft	0 to 0.25	0 to 25
3-5	Soft	0.25 to 0.50	25 to 50
6-9	Medium	0.50 to 1	50 to 100
10-16	Stiff	1 to 2	100 to 200
17-30	Very stiff	2 to 4	200 to 400
>30	Hard	>4	>400

Table 3: N and ϕ Relation to Relative Density for Fine Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
1 – 2	Very Loose	0 – 15	26 – 28	70 – 100
3 – 6	Loose	15 – 35	28 – 30	90 – 115
7 – 15	Medium	35 – 65	30 – 34	110 – 130
16 – 30	Dense	65 – 85	33 – 38	110 – 140
>30	Very Dense	>85	>50	130 – 150

Table 4: N and ϕ Relation to Relative Density for Medium Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
2 – 3	Very Loose	0 – 15	27 – 28	70 – 100
4 – 7	Loose	15 – 35	30 – 32	90 – 115
8 – 20	Medium	35 – 65	32 – 36	110 – 130
21 – 40	Dense	65 – 85	36 – 42	110 – 140

Table 5: N and ϕ Relation to Relative Density for Coarse Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
3 – 6	Very Loose	0 – 15	28 – 30	70 – 100
5 – 9	Loose	15 – 35	30 – 34	90 – 115
10 – 25	Medium	35 – 65	33 – 40	110 – 130
26 – 40	Dense	65 – 85	40 – 50	110 – 140
>40	Very Dense	>85	>50	130 – 150

Bearing capacity for deep foundation

Foundations, preferably piles are considered to transmit heavy load from superstructure to deep and hard strata. As the formation of soil strata consists of cohesive and non-cohesive type, the evaluation criteria for the determination of skin friction and the end bearing capacities of piles also vary. Details of the evaluation are provided in the following sub-headings.

Table 6: Formula Used to Determine Bearing Capacity of Soil

Shallow Foundation	
For Cohesive Soil	
$q_a = q_u + \gamma \frac{D}{3}$ $q_a = q_u$(Common Practice)	Where, q_a=Allowable bearing capacity, q_u= Unconfined Compressive Strength, γ= Unit Weight of soil, D= Depth of footing
For Cohesion less Soil	
$q_{ult} = 30N;$ $q_a = q_{ult}/FS$	q_{ult}= Ultimate bearing capacity in kPa, N = SPT Number, FS= Factor of Safety
Source: Bowles, J. E., "Physical and Geotechnical Properties of Soils", 2nd Edition	
Deep Foundation	
For Cohesive Soil	
$f_{su} = \alpha C_u$; $q_{pu} = N_c C_u$	α = Adhesion factor, f_{su} = Ultimate skin friction, C_u = Undrained Shear Strength, q_{pu} = Ultimate end bearing of pile, N_c = Bearing capacity factor for deep foundation=9
For Cohesion less Soil	
$f_{su} = N/50\ tsf$; $q_{pu} = 4N\ tsf$	f_{su} = Ultimate skin friction, q_{pu} = Ultimate end bearing of pile, N = SPT Number
Source: Rajapakse RA (2008), "Geotechnical Engineering Calculations and Rules of Thumb"	

Based on the above formula, the important parameters for foundation design have been determined and are given in **Table 7**, **Table 8, Table 9, Table 10 and Table 11** respectively.

Table 7: The allowable bearing capacity for shallow foundation in Tsf (FS=2.5)

Bore Hole No.	Depth	Design N	Clay						Sand
			Continuous			Square(Isolated)			
	(ft)	-	D/B=0.25	D/B=0.5	D/B=1	D/B=0.25	D/B=0.5	D/B=1	
BH01	2.5-5.0	1	0.162	0.171	0.186	0.192	0.207	0.225	0.175
	5.0-10	3	0.316	0.332	0.362	0.373	0.403	0.439	0.341
BH02	2.5-5.0	2	0.278	0.293	0.319	0.329	0.355	0.386	0.300
	5.0-10	3	0.352	0.370	0.404	0.416	0.449	0.489	0.379
BH03	2.5-5.0	2	0.255	0.268	0.293	0.301	0.326	0.354	0.275
	5.0-10	3	0.352	0.370	0.404	0.416	0.449	0.489	0.379
BH04	2.5-5.0	2	0.232	0.244	0.266	0.274	0.296	0.322	0.250
	5.0-10	3	0.352	0.370	0.404	0.416	0.449	0.489	0.379
BH05	2.5-5.0	2	0.278	0.293	0.319	0.329	0.355	0.386	0.300
	5.0-10	3	0.396	0.416	0.454	0.468	0.505	0.550	0.427
BH06	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH07	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH08	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH09	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH10	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-

Table 8: The allowable skin friction for Precast piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.025	0.049	0.025	0.049	0.049	-	-	-	-	-
10	0.025	0.049	0.049	0.049	0.025	-	-	-	-	-
15	0.049	0.074	0.074	0.049	0.096	-	-	-	-	-
20	0.118	0.096	0.096	0.118	0.118	-	-	-	-	-
25	0.138	0.118	0.064	0.096	0.096	-	-	-	-	-
30	0.072	0.074	0.064	0.064	0.156	-	-	-	-	-
35	0.112	0.174	0.088	0.072	0.080	-	-	-	-	-
40	0.096	0.120	0.112	0.128	0.128	-	-	-	-	-
45	0.120	0.104	0.136	0.144	0.136	-	-	-	-	-
50	0.096	0.128	0.136	0.128	0.128	-	-	-	-	-
55	0.104	0.192	0.168	0.184	0.160	-	-	-	-	-
60	0.184	0.216	0.184	0.184	0.176	-	-	-	-	-
65	0.184	0.240	0.200	0.232	0.184	-	-	-	-	-
70	0.152	0.208	0.160	0.152	0.192	-	-	-	-	-
75	0.184	0.208	0.128	0.184	0.192	-	-	-	-	-
80	0.176	0.168	0.176	0.192	0.168	-	-	-	-	-
85	0.200	0.176	0.184	0.168	0.176	-	-	-	-	-
90	0.200	0.184	0.184	0.160	0.176	-	-	-	-	-
95	0.160	0.176	0.184	0.168	0.152	-	-	-	-	-
100	0.168	0.160	0.152	0.168	0.128	-	-	-	-	-
105	0.176	0.168	0.096	0.064	0.032	-	-	-	-	-
110	0.152	0.144	0.072	0.040	0.032	-	-	-	-	-
115	0.168	0.128	0.048	0.048	0.064	-	-	-	-	-
120	0.136	0.024	0.056	0.064	0.048	-	-	-	-	-
125	0.064	0.104	0.152	0.136	0.144	-	-	-	-	-
130	0.136	0.160	0.160	0.144	0.152	-	-	-	-	-
135	0.144	0.144	0.136	0.144	0.152	-	-	-	-	-

Table 9: The allowable end bearing for Precast piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.225	0.450	0.225	0.450	0.450	-	-	-	-	-
10	0.225	0.450	0.450	0.450	0.225	-	-	-	-	-
15	0.450	0.675	0.675	0.450	0.900	-	-	-	-	-
20	1.125	0.900	0.900	1.125	1.125	-	-	-	-	-
25	1.350	1.125	12.800	0.900	0.900	-	-	-	-	-
30	14.400	0.675	12.800	12.800	1.575	-	-	-	-	-
35	22.400	1.800	17.600	14.400	16.000	-	-	-	-	-
40	19.200	24.000	22.400	25.600	25.600	-	-	-	-	-
45	24.000	20.800	27.200	28.800	27.200	-	-	-	-	-
50	19.200	25.600	27.200	25.600	25.600	-	-	-	-	-
55	20.800	38.400	33.600	36.800	32.000	-	-	-	-	-
60	36.800	43.200	36.800	36.800	35.200	-	-	-	-	-

65	36.800	48.000	40.000	46.400	36.800	-	-	-	-	-
70	30.400	41.600	32.000	30.400	38.400	-	-	-	-	-
75	36.800	41.600	25.600	36.800	38.400	-	-	-	-	-
80	35.200	33.600	35.200	38.400	33.600	-	-	-	-	-
85	40.000	35.200	36.800	33.600	35.200	-	-	-	-	-
90	40.000	36.800	36.800	32.000	35.200	-	-	-	-	-
95	32.000	35.200	36.800	33.600	30.400	-	-	-	-	-
100	33.600	32.000	30.400	33.600	25.600	-	-	-	-	-
105	35.200	33.600	19.200	12.800	6.400	-	-	-	-	-
110	30.400	28.800	14.400	8.000	6.400	-	-	-	-	-
115	33.600	25.600	9.600	9.600	12.800	-	-	-	-	-
120	27.200	4.800	11.200	12.800	9.600	-	-	-	-	-
125	12.800	20.800	30.400	27.200	28.800	-	-	-	-	-
130	27.200	32.000	32.000	28.800	30.400	-	-	-	-	-
135	28.800	28.800	27.200	28.800	30.400	-	-	-	-	-

Table 10: The allowable skin friction for Bored piles in Tsf (FS=2.5)

Depth	**Bore Hole**									
ft	**BH01**	**BH02**	**BH03**	**BH04**	**BH05**	**BH06**	**BH07**	**BH08**	**BH09**	**BH10**
5	0.016	0.032	0.016	0.032	0.032	-	-	-	-	-
10	0.016	0.032	0.032	0.032	0.016	-	-	-	-	-
15	0.032	0.048	0.048	0.032	0.062	-	-	-	-	-
20	0.077	0.062	0.062	0.077	0.077	-	-	-	-	-
25	0.090	0.077	0.042	0.062	0.062	-	-	-	-	-
30	0.047	0.048	0.042	0.042	0.101	-	-	-	-	-
35	0.073	0.113	0.057	0.047	0.052	-	-	-	-	-
40	0.062	0.078	0.073	0.083	0.083	-	-	-	-	-
45	0.078	0.068	0.088	0.094	0.088	-	-	-	-	-
50	0.062	0.083	0.088	0.083	0.083	-	-	-	-	-
55	0.068	0.125	0.109	0.120	0.104	-	-	-	-	-
60	0.120	0.140	0.120	0.120	0.114	-	-	-	-	-
65	0.120	0.156	0.130	0.151	0.120	-	-	-	-	-
70	0.099	0.135	0.104	0.099	0.125	-	-	-	-	-
75	0.120	0.135	0.083	0.120	0.125	-	-	-	-	-
80	0.114	0.109	0.114	0.125	0.109	-	-	-	-	-
85	0.130	0.114	0.120	0.109	0.114	-	-	-	-	-
90	0.130	0.120	0.120	0.104	0.114	-	-	-	-	-
95	0.104	0.114	0.120	0.109	0.099	-	-	-	-	-
100	0.109	0.104	0.099	0.109	0.083	-	-	-	-	-
105	0.114	0.109	0.062	0.042	0.021	-	-	-	-	-
110	0.099	0.094	0.047	0.026	0.021	-	-	-	-	-
115	0.109	0.083	0.031	0.031	0.042	-	-	-	-	-
120	0.088	0.016	0.036	0.042	0.031	-	-	-	-	-
125	0.042	0.068	0.099	0.088	0.094	-	-	-	-	-
130	0.088	0.104	0.104	0.094	0.099	-	-	-	-	-
135	0.094	0.094	0.088	0.094	0.099	-	-	-	-	-

Table 11: The allowable end bearing for Bored piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.101	0.203	0.101	0.203	0.203	-	-	-	-	-
10	0.101	0.203	0.203	0.203	0.101	-	-	-	-	-
15	0.203	0.304	0.304	0.203	0.405	-	-	-	-	-
20	0.506	0.405	0.405	0.506	0.506	-	-	-	-	-
25	0.608	0.506	5.760	0.405	0.405	-	-	-	-	-
30	6.480	0.304	5.760	5.760	0.709	-	-	-	-	-
35	10.080	0.810	7.920	6.480	7.200	-	-	-	-	-
40	8.640	10.800	10.080	11.520	11.520	-	-	-	-	-
45	10.800	9.360	12.240	12.960	12.240	-	-	-	-	-
50	8.640	11.520	12.240	11.520	11.520	-	-	-	-	-
55	9.360	17.280	15.120	16.560	14.400	-	-	-	-	-
60	16.560	19.440	16.560	16.560	15.840	-	-	-	-	-
65	16.560	21.600	18.000	20.880	16.560	-	-	-	-	-
70	13.680	18.720	14.400	13.680	17.280	-	-	-	-	-
75	16.560	18.720	11.520	16.560	17.280	-	-	-	-	-
80	15.840	15.120	15.840	17.280	15.120	-	-	-	-	-
85	18.000	15.840	16.560	15.120	15.840	-	-	-	-	-
90	18.000	16.560	16.560	14.400	15.840	-	-	-	-	-
95	14.400	15.840	16.560	15.120	13.680	-	-	-	-	-
100	15.120	14.400	13.680	15.120	11.520	-	-	-	-	-
105	15.840	15.120	8.640	5.760	2.880	-	-	-	-	-
110	13.680	12.960	6.480	3.600	2.880	-	-	-	-	-
115	15.120	11.520	4.320	4.320	5.760	-	-	-	-	-
120	12.240	2.160	5.040	5.760	4.320	-	-	-	-	-
125	5.760	9.360	13.680	12.240	12.960	-	-	-	-	-
130	12.240	14.400	14.400	12.960	13.680	-	-	-	-	-
135	12.960	12.960	12.240	12.960	13.680	-	-	-	-	-

CONCLUSION AND RECOMMENDATION

On the basis of analysis and information provided by the owner, following recommendations are suggested for the proposed ten (10) storied commercial building.

Soil profile scenario

Overall soil information of the investigated site is quite irregular in between the boreholes and remarkable vertical inconsistency exists in the soil formation. Nearly two types of scenarios have been observed among the boreholes.

For type A (borehole no.: 01, 02, 04 & 05)- gray soft clay with poor bearing capacity exists roughly up to the top 25- 35 ft depth below EGL followed by gray fine sand with silt and clay layer with moderate bearing capacity up to 25–60 ft depth. Though gray fine sand having good bearing capacity has been found up to 55–120 ft depth below EGL among the boreholes but a significant declination of bearing capacity has been observed due to the presence of gray fine sand with silt and clay layer. Finally, gray fine sand with good bearing capacity initiates from 125-130 ft of EGL in those boreholes and continues thereafter.

For type B (borehole no.: 03)- gray soft clay with poor bearing capacity exists roughly up to the top 20 ft from EGL. After that gray fine sand with silt and clay with moderate bearing capacity has been observed followed by gray fine sand having with good bearing capacity. The next gray fine sand with silt and clay, and gray fine sand layer have been found 75-120 ft depth below EGL but declination of bearing capacity has been noticed due to the presence of gray fine sand with silt and clay layer in the deeper portion. Finally, a gray fine sand layer with good bearing capacity initiates from 125 ft of EGL in the borehole and continues to a greater depth.

From the above discussion it is very clear that the whole scenario of the proposed site is quite different in nature.

Foundation options

In this situation, the foundation should be selected considering both the dimension and design load for the proposed building. Shallow foundations may not be a suitable option for ten storied commercial building as soft clay soil with very poor bearing capacity exists at the top in most of the boreholes. Although, bearing capacity values for shallow foundations can be found from the attached **Table 7** in the report which is based on existing soil conditions.

However, deep foundation may be preferentially recommended to ensure the stability of the structure. Size and depth of the pile should be selected in the light of bearing capacity given in the report.

Safety provisions

Since the proposed site is composed of alternate layers of soft cohesive and granular soil, it is reasonably vulnerable to differential settlement. Therefore, it is strongly recommended to check the **settlement** of the site considering building load while designing either shallow or deep foundation. Obviously, **pile load test** covering the entire project area must be carried out for confirming the design parameters of the service pile (*if pile foundation is considered*). Moreover, the earthquake vulnerability of the project area as well as the quality management of the construction work including workmanship should be taken into account for the safe design of shallow or deep foundation.

Attachments

The layout of the bore holes of the investigated site and the bore-log profiles with SPT values is shown in the Appendix. Client performed no special laboratory test for the proposed site.

Special notes

These suggestions are absolutely tentative and selection of type, depth and dimensions of foundation would be decided by the Consultant or Design Engineer. It is also to be noted that the above recommendation is restricted only to the existing soil condition of **135** ft as client provided.

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